

Applied Wave Optics: Maxwell Equations

Reinhard Caspary

Version date: May 20, 2026



Except where otherwise noted, this document and its content are licensed under the Creative Commons Attribution-ShareAlike 4.0 International license.

All sources and vector graphics are available on the GitHub repository
https://github.com/reincas/AW0_course.

Maxwell's equations

The most general differential form of Maxwell's equations is

$$\nabla \mathbf{D} = \rho \quad \text{Gauss's law}$$

$$\nabla \mathbf{B} = 0 \quad \text{Gauss's law for magnetism}$$

$$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t \quad \text{Faraday's law}$$

$$\nabla \times \mathbf{H} = \mathbf{j} + \partial \mathbf{D} / \partial t \quad \text{Ampère-Maxwell law}$$

with the electric field vector $\mathbf{E}$ [V/m] and its material dependent form, the displacement field $\mathbf{D}$ [As/m²] as well as the magnetic field vector $\mathbf{H}$ [A/m] and its material dependent form, the magnetic induction $\mathbf{B}$ [Vs/m²]. The scalar electrical charge density is ρ [C/m³] and the local flux of electrical charges is taken into account by the current density vector $\mathbf{j}$ [A/m²].

Polarisation

The material dependent relationship between $\mathbf{D}$ and $\mathbf{E}$ is

$$\mathbf{D}(\mathbf{E}) = \varepsilon_0 \mathbf{E} + \mathbf{P}(\mathbf{E})$$

with the permittivity $\varepsilon_0 = 8.854 \cdot 10^{-12} \frac{\text{As}}{\text{Vm}}$ and the polarisation vector $\mathbf{P}$. Its Taylor expansion gives

$$\mathbf{P}(\mathbf{E})|_{\mathbf{E}=0} = \mathbf{P}_0 + \varepsilon_0 \chi_e^{(1)} \mathbf{E} + \varepsilon_0 \chi_e^{(2)} \mathbf{E}^2 + \dots$$

where the first term is a **static** polarisation, characteristic for ferroelectric materials, the second term describes the **linear** behaviour and all following terms **non-linear** dependencies with the electric susceptibility tensors $\chi_e^{(n)}$.

Magnetisation

The material dependent relationship between $\mathbf{H}$ and $\mathbf{B}$ is

$$\mathbf{B}(\mathbf{H}) = \mu_0 \mathbf{H} + \mu_0 \mathbf{M}(\mathbf{H})$$

with the permeability $\mu_0 = 4\pi \cdot 10^{-7} \frac{\text{Vs}}{\text{Am}}$ and the magnetisation vector $\mathbf{M}$. Its Taylor expansion gives

$$\mathbf{M}(\mathbf{H})|_{\mathbf{H}=0} = \mathbf{M}_0 + \chi_m^{(1)} \mathbf{H} + \chi_m^{(2)} \mathbf{H}^2 + \dots$$

where the first term is a **static** magnetisation, characteristic for ferromagnetic materials, the second term describes the **linear** behaviour and all following terms **non-linear** dependencies with the magnetic susceptibility tensors $\chi_m^{(n)}$.

Linear Case

For isotropic materials, all susceptibilities are scalars and the linear cases simplify to

$$\mathbf{D} = \varepsilon \varepsilon_0 \mathbf{E}$$

$$\mathbf{B} = \mu \mu_0 \mathbf{H}$$

with the two dimensionless scalar quantities called **relative permittivity** $\varepsilon = 1 + \chi_e^{(1)}$ and **relative permeability** $\mu = 1 + \chi_m^{(1)}$.

Note

We always assume this case in the following.

Continuity Equation

Identity relation from vector algebra for an arbitrary field $\mathbf{A}$:

$$\nabla(\nabla \times \mathbf{A}) = 0$$

It must therefore also be true for the magnetic field and we can utilize Maxwell's equations:

$$\begin{aligned}\nabla(\nabla \times \mathbf{H}) &= \nabla \left(\mathbf{j} + \frac{\partial \mathbf{D}}{\partial t} \right) \\ &= \nabla \mathbf{j} + \frac{\partial}{\partial t} \nabla \mathbf{D} \\ &= \nabla \mathbf{j} + \frac{\partial \rho}{\partial t} \mathbf{1} \stackrel{!}{=} 0 \\ \nabla \mathbf{j} &= -\frac{\partial \rho}{\partial t}\end{aligned}$$

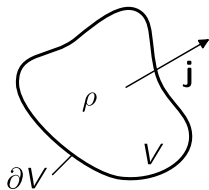
This is the continuity equation for electrical charges in its differential (local) form.

Continuity Equation (cont.)

The meaning of the continuity equation becomes more obvious, when it is integrated:

$$\int_{\partial V} \mathbf{j} d\mathbf{A} = -\frac{dQ}{dt}$$

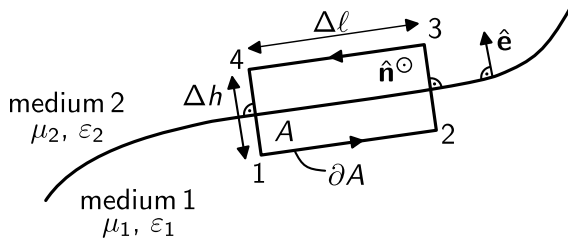
The left hand side is the total electrical current through the surface of a volume V and the right hand side the temporal variation of the total electrical charge $Q = \int \rho dV$ inside this volume.



Maxwell's equations guarantee that both quantities are identical.

Parallel Fields at Interfaces

Interface between two different materials:



Differentials for a line integration along the path ∂A :

$$1 \rightarrow 2 : \quad d\mathbf{r} = -\hat{n} \times \hat{e} \, dx$$

$$2 \rightarrow 3 : \quad d\mathbf{r} = \hat{e} \, dx$$

$$3 \rightarrow 4 : \quad d\mathbf{r} = \hat{n} \times \hat{e} \, dx$$

$$4 \rightarrow 1 : \quad d\mathbf{r} = -\hat{e} \, dx$$

Parallel Fields at Interfaces (cont.)

Stokes theorem from vector analysis:

$$\int_A (\nabla \times \mathbf{E}) d\mathbf{A} = \oint_{\partial A} \mathbf{E} d\mathbf{r}$$

We shrink the area $A \rightarrow 0$, and use the fact that for $\Delta\ell \rightarrow 0$ the line integrals $2 \rightarrow 3$ and $4 \rightarrow 1$ are identical, but with opposite sign:

$$\lim_{A \rightarrow 0} \int_A (\nabla \times \mathbf{E}) d\mathbf{A} = \lim_{\Delta h, \Delta\ell \rightarrow 0} \left[- \int_1^2 (\hat{\mathbf{n}} \times \hat{\mathbf{e}}) \mathbf{E}_1 dx + \int_3^4 (\hat{\mathbf{n}} \times \hat{\mathbf{e}}) \mathbf{E}_2 dx \right]$$

Now we insert Faraday's law on the left and on the right the scalar triple products extract the field components parallel to the interface:

$$- \lim_{A \rightarrow 0} \frac{\partial}{\partial t} \int_A \mathbf{B} d\mathbf{A} = \lim_{\Delta h, \Delta\ell \rightarrow 0} \left[- \int_1^2 E_{1\parallel} dx + \int_3^4 E_{2\parallel} dx \right]$$

Parallel Fields at Interfaces (cont.)

The left side is obviously zero and on the right side we use the fact that for $\Delta\ell \rightarrow 0$ the fields can be treated as constant:

$$0 = (E_{1\parallel} - E_{2\parallel}) \Delta\ell$$

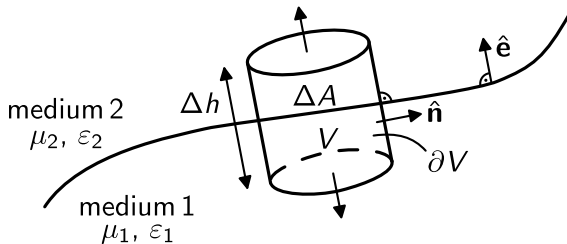
We get the same result when we carry out this calculation for the magnetic field and thus end with the following boundary conditions for field components parallel to the interface plane:

$$E_{1\parallel} = E_{2\parallel}$$

$$H_{1\parallel} = H_{2\parallel}$$

Normal Fields at Interfaces

Interface between two different materials:



Differentials for an integration over the surface ∂V :

$$\text{top base : } d\mathbf{A} = \hat{e} dA$$

$$\text{lateral : } d\mathbf{A} = \hat{n} dA$$

$$\text{bottom base : } d\mathbf{A} = -\hat{e} dA$$

Normal Fields at Interfaces (cont.)

Gauss's theorem from vector analysis:

$$\int_V \nabla \cdot \mathbf{D} \, dV = \oint_{\partial V} \mathbf{D} \cdot d\mathbf{A}$$

We shrink the volume $V \rightarrow 0$, and use the fact that in this case the lateral surface integrals in both materials vanish, because the field is constant:

$$\lim_{V \rightarrow 0} \int_V \nabla \cdot \mathbf{D} \, dV = \lim_{\Delta A, \Delta h \rightarrow 0} \left[\int_{\Delta A_1} \mathbf{D}_1 \cdot \hat{\mathbf{e}} \, dA - \int_{\Delta A_2} \mathbf{D}_2 \cdot \hat{\mathbf{e}} \, dA \right]$$

Now we insert Gauss's law on the left and on the right the scalar products extract the component of the displacement field perpendicular to the interface:

$$\lim_{V \rightarrow 0} \int_V \rho \, dV = \lim_{\Delta A, \Delta h \rightarrow 0} \left[\int_{\Delta A} D_{1\perp} \, dA - \int_{\Delta A} D_{2\perp} \, dA \right]$$

Normal Fields at Interfaces (cont.)

The left side is obviously zero and on the right side we use the fact that for $\Delta A \rightarrow 0$ the fields can be treated as constant:

$$0 = (D_{1\perp} - D_{2\perp}) \Delta A$$

We get the same result when we carry out this calculation for the magnetic induction and thus end with the following boundary conditions for field components perpendicular to the interface plane:

$$\begin{aligned} D_{1\perp} &= D_{2\perp} & \varepsilon_1 E_{1\perp} &= \varepsilon_2 E_{2\perp} \\ B_{1\perp} &= B_{2\perp} & \mu_1 H_{1\perp} &= \mu_2 H_{2\perp} \end{aligned}$$